

Generative AI: The Challenges and Some Solutions



Generative AI is revolutionary but poses a few challenges too. However, there are quite a few solutions that are being developed to improve its effectiveness, and reduce inherent biases and copyright infringements. Smaller companies and startups will benefit from these.

Generative AI, epitomised by ChatGPT, is not a sudden phenomenon but a result of sustained efforts bolstered by the contributions of the open source community. The collaborative environment has been pivotal in realising practical applications of generative AI, from coding assistance and email composition to fostering creativity by generating new code or data.

Scope of generative AI in India: Addressing challenges and solutions

At a recent conference on open source, industry leaders and gurus explored pressing questions about the evolution of generative AI. Dr Pushpak Bhattacharyya, professor of computer science and engineering at IIT Bombay, said: “Generative AI aligns with the government of India’s Digital

India initiative. A machine translation project called *Bhashini*, which uses generative AI, aims to translate from all Indian languages to all other Indian languages. It has applications in education, the judicial system, agricultural domain, and so on.”

Challenges and pitfalls of generative AI

Generative AI poses challenges inherent in its probabilistic nature, carrying biases and errors due to imperfect training

Horizons of generative AI in the Indian context

- Generative AI permeates various sectors in India, enhancing user experiences in content creation, drafting emails, and document summarisation.
- In the legal sector, generative AI aids in summarising large volumes of documents, analysing cases, and improving the efficiency of legal professionals.
- The agricultural industry benefits from AI advisory systems that analyse weather and soil data to provide recommendations for farming activities.
- To maximise the benefits of AI, developing user-friendly interfaces that support India’s many regional languages is crucial, leveraging advances in machine translation and NLP for effective communication.
- These efforts align with the Digital India campaign’s goals to create a digitally empowered society.